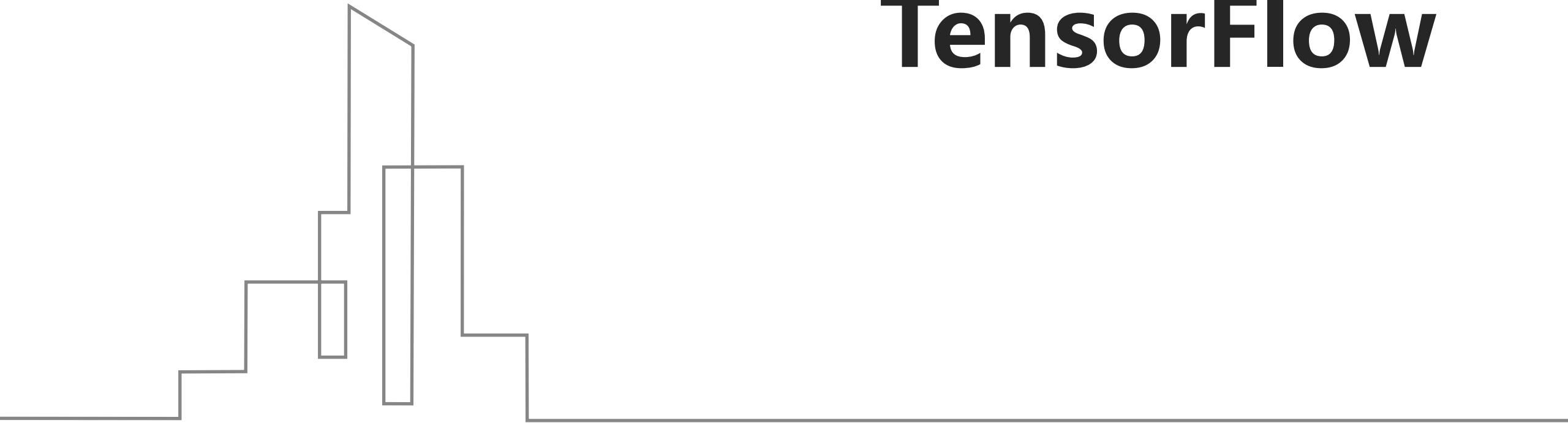


TensorFlow



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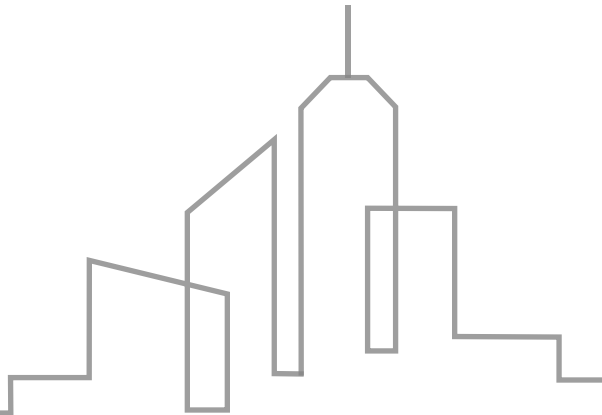
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01

Introduction: Foundational Choice

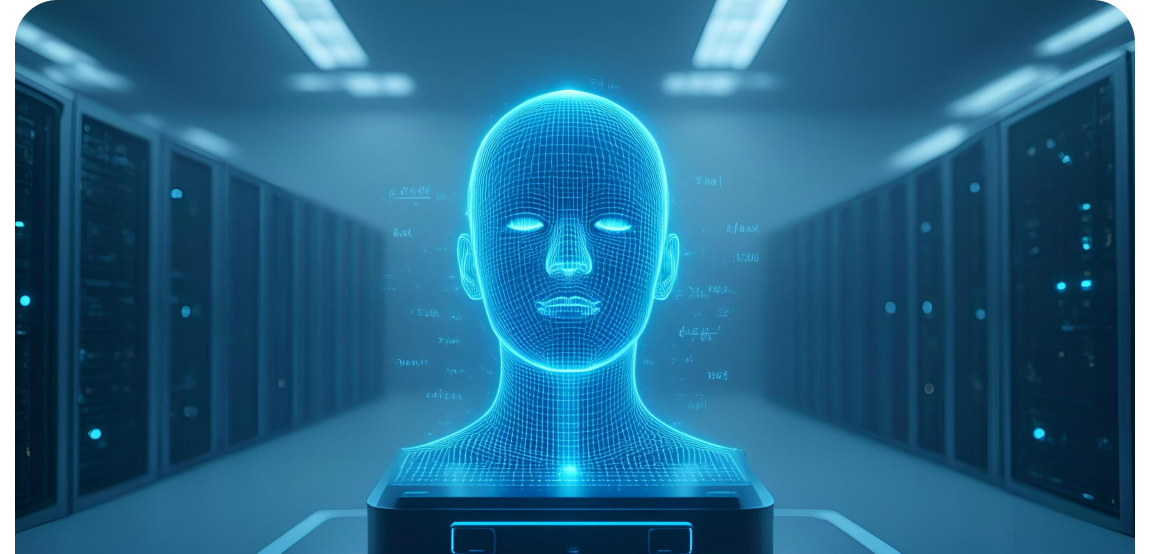


Core goal: Scalable AI models



Scalable AI models

Building AI models that can handle large-scale data and complex tasks efficiently



Production-ready models

Ensuring models are robust, reliable, and ready for deployment in real-world applications



What is TensorFlow?

Google's ML platform

Google's ML platform

TensorFlow is a comprehensive, open-source machine learning platform developed by Google

Complete ML platform

It offers a range of tools for building and deploying machine learning models at scale



Relevance to saascv: Core framework for CV solutions



Core framework for CV

TensorFlow is fundamental to our computer vision solutions, providing the necessary tools and infrastructure

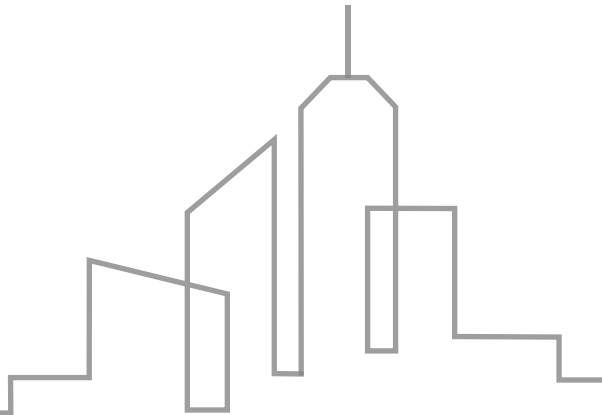
Large-scale deployment

It enables the deployment of CV models across various platforms and devices, ensuring scalability and efficiency



02

Core Concepts: Two Sides of TensorFlow



High-Level API: Keras for simplicity



Simplicity and rapid prototyping

Keras, as a high-level API, enables simple and fast prototyping with TensorFlow.

Sequential vs. Functional APIs

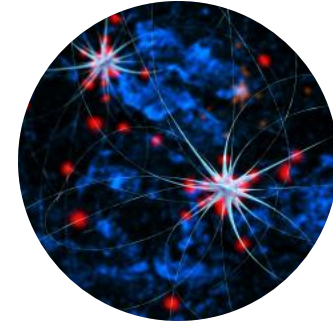
Keras offers both Sequential and Functional APIs, providing flexibility for model creation.

Sequential vs. Functional APIs



Choice between API styles

Sequential API is straightforward for simple models, while Functional API offers more flexibility for complex architectures.



Flexibility in model design

Functional API allows for more complex model designs, including models with multiple inputs and outputs.



Low-Level Core: TensorFlow Core for control



Fine-grained control

TensorFlow Core provides detailed control over the model, essential for advanced customization and optimization.



Multi-dimensional arrays

TensorFlow Core operates on tensors, which are multi-dimensional arrays, fundamental for data representation.



Fundamental data structure: Tensors



Multi-dimensional arrays

Tensors are the fundamental data structure in TensorFlow, representing multi-dimensional arrays of data.



Essential for computations

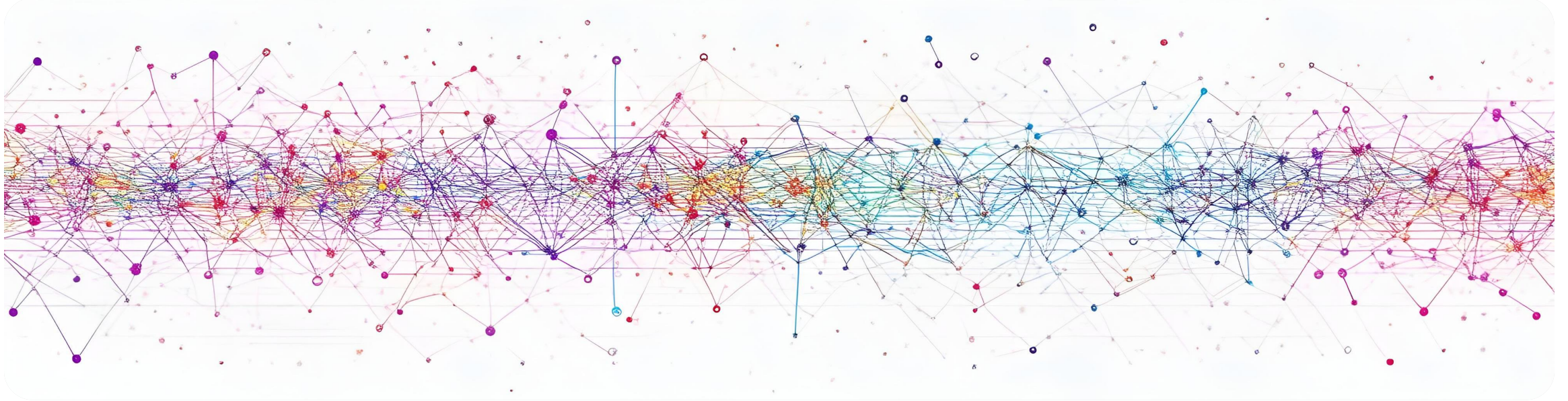
Tensors are used to perform computations and are the building blocks for creating scalable AI models.

03

Architecture: Evolution from Static to Eager



Past: TF 1.x with static computation graphs



Static Computation Graphs

TF 1.x used static graphs, which required defining the model structure before running it.

Debugging Nightmare

Debugging was complex due to the lack of immediate feedback during development.



Debugging and optimization challenges

Optimization Trade-offs

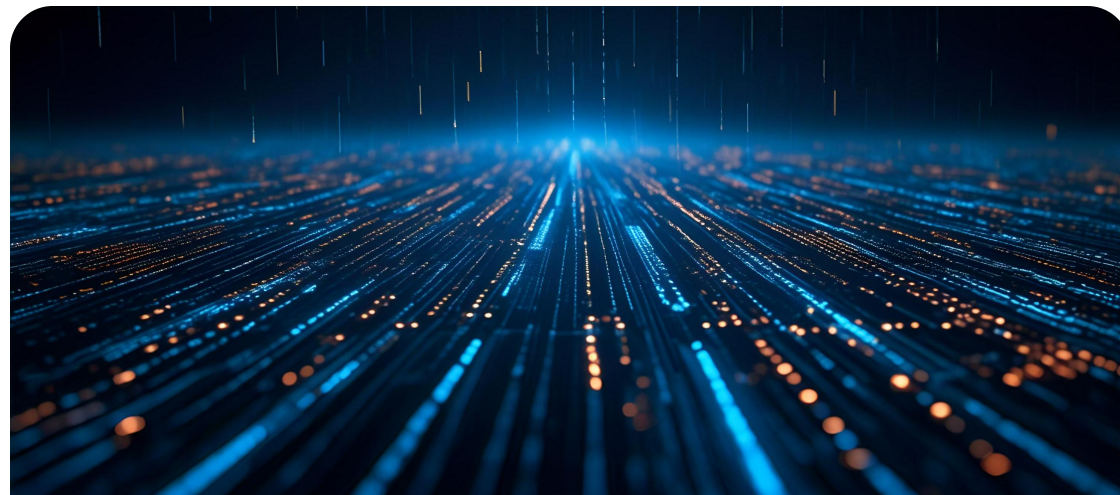
Static graphs made optimization challenging as it required a deep understanding of the entire graph.

Present: TF 2.x with eager execution



Eager Execution

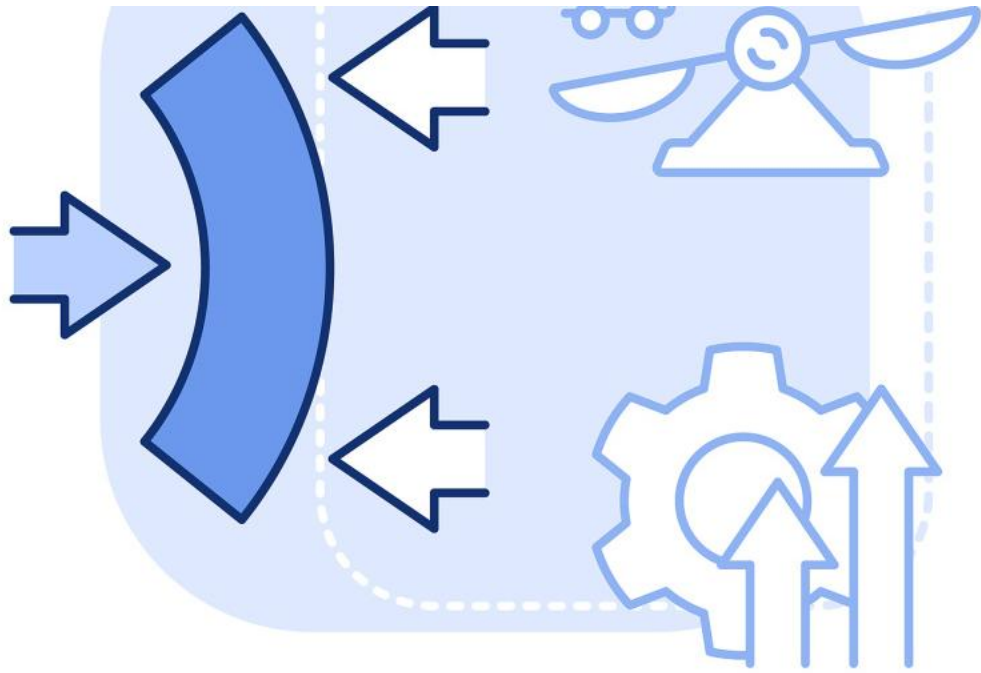
TF 2.x introduced eager execution, allowing operations to be computed immediately.



Define-by-Run Paradigm

This paradigm simplifies model development by defining and running operations on the fly.

Benefits: Easier debugging, intuitive flow, speed



Increase Flexibility

Easier Debugging

Eager execution provides immediate feedback, making debugging more straightforward.

Intuitive Python Flow

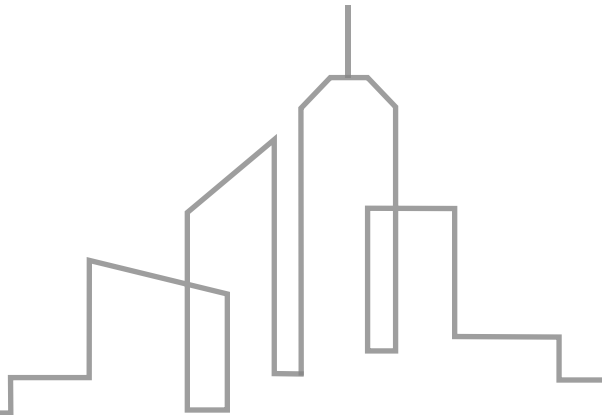
Developers can use intuitive Python control flow, which is more natural and easier to understand.

Model Iteration Speed

Immediate execution speeds up the iteration process, allowing for faster model development.

04

Ecosystem: Production-Grade Tools



TensorBoard for model monitoring



Model health and training progress

TensorBoard monitors the health of models and the progress of training, providing insights into performance metrics and visualizations.



Essential for debugging and optimization

It is crucial for debugging models and optimizing performance, offering a comprehensive view of the training process.

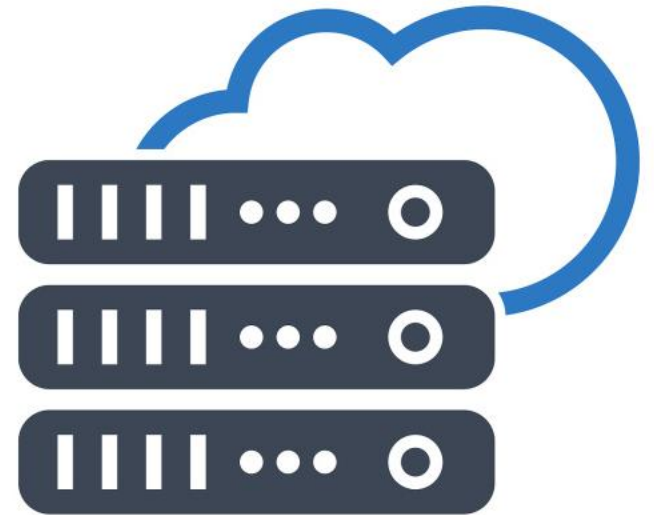
tf.data & TFDS for data pipelines

High-performance data input pipelines

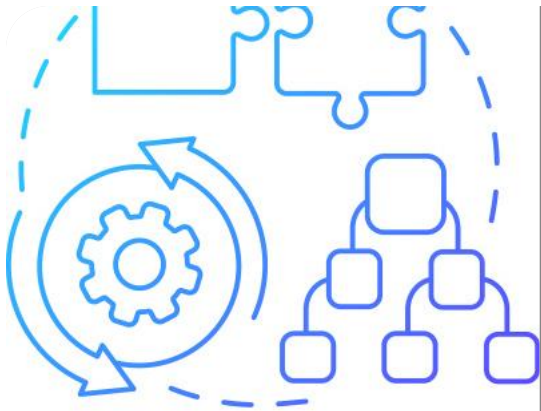
tf.data and TFDS are essential for constructing efficient data pipelines, particularly for large computer vision datasets.

Critical for large CV datasets

These tools handle the complexities of large-scale data, ensuring smooth and scalable data processing for CV tasks.

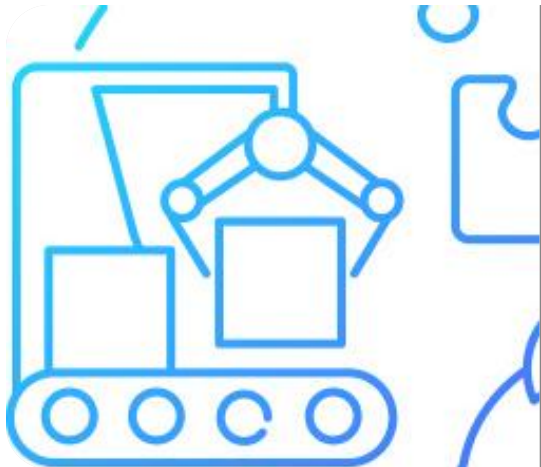


TensorFlow Extended (TFX) for MLOps



MLOps framework for production

TFX is the MLOps framework that enables the production and consistency of TensorFlow models in real-world applications.

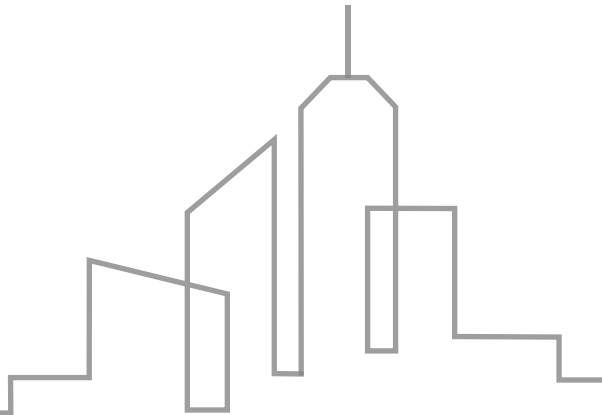


Ensuring production and consistency

It provides a complete set of tools for managing the machine learning workflow, from data preparation to model deployment.

05

Deployment: SaaSv Models to the World





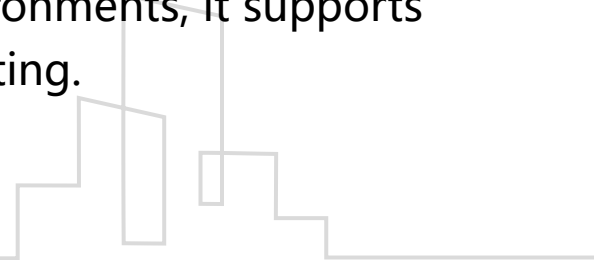
TensorFlow Serving for cloud inference

High-performance cloud inference

TensorFlow Serving enables scalable, high-performance model serving in the cloud.

Scalable architecture

Designed for production environments, it supports model versioning and A/B testing.



TensorFlow Lite for mobile and edge devices

Mobile and edge deployment

TensorFlow Lite allows for deploying models on mobile and edge devices for client-side processing.

Efficient model optimization

It includes model optimization techniques to reduce size and increase speed on devices.



TensorFlow.js for web browser deployment

Web browser CV models

TensorFlow.js enables running computer vision models directly in the web browser, facilitating demos and prototyping.

JavaScript integration

It allows for seamless integration with JavaScript, making it accessible for web developers.



Conclusion: End-to-end solution for saascv production



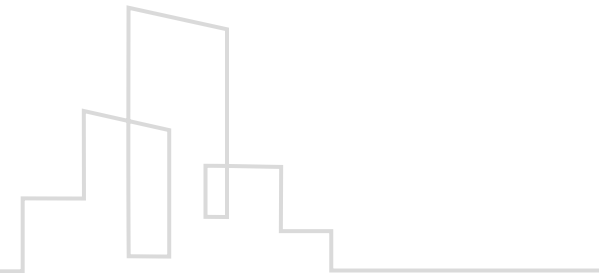
Complete end-to-end solution

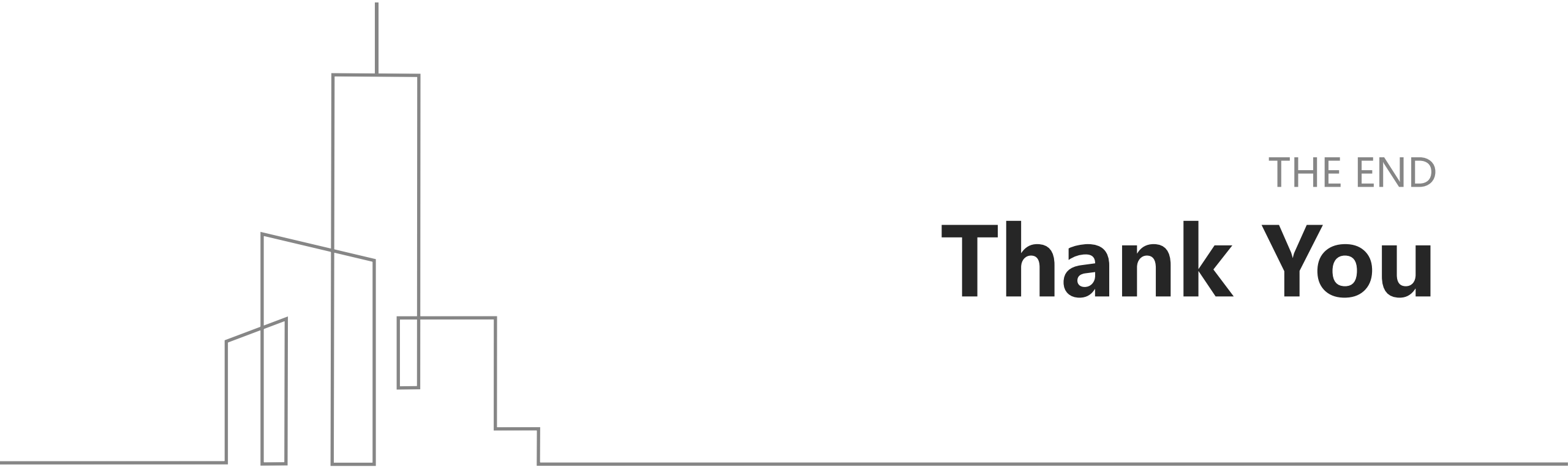
TensorFlow provides a comprehensive, end-to-end solution for building, deploying, and maintaining saascv production models.



Production-grade tools

With TensorFlow Extended (TFX) and other ecosystem tools, it ensures production consistency and scalability.





THE END

Thank You